

DUAN Zihang Raymond

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MSc in Information Technology student and aspiring Software Engineer, passionate about AI, IoT, and Full-Stack Software Development.

Education

Hong Kong University of Science and Technology (HKUST)

Sep 2026 – Jun 2027 (Expected)

MSc in Information Technology

Coursework: AI & Machine Learning, Data Mining & Knowledge Discovery, Reliable Software Systems, Principles of Cybersecurity

Hong Kong University of Science and Technology (HKUST)

Sep 2022 – Jun 2026

BEng in Computer Engineering — Second Class Honors, Division 1

Coursework: AI & Vision, Systems & Networks, Software Engineering, Architecture & Multimedia

Work Experience

EFSPost

Jan 2026 – May 2026

IT Specialist (Part-time)

Remote

- **Customer Service Automation:** Integrated inquiry workflows into NetEase Qiyu (knowledge base + AI chatbot), easing customer service workload.
- **System Integration & Tuning:** Connected systems to Qiyu with AI/LLM first-line responses and automatic escalation to human agents for complex cases.
- **Multi-Channel Consolidation:** Unified multiple WeChat CS accounts into one Qiyu platform, reducing monitoring burden.

Growgreen Limited

Jun 2026 – Jul 2026

Software Engineer Intern (Full-time)

Science Park, Hong Kong

- **Product Analysis:** Studied company hardware design, component architecture, and integration requirements.
- **Website Migration:** Migrated the website from WordPress to Shopify; built storefront pages using Liquid and CSS.
- **Modular Knowledge Architecture:** Built a modular Markdown knowledge hierarchy on Alibaba Cloud AgentRun for path-based context routing and improved retrieval accuracy.
- **Efficient Context Management:** Reduced token usage via on-demand, skill-by-skill dynamic loading instead of monolithic skill structures.
- **System Prompt & Persona Engineering:** Crafted system prompts and brand persona with safeguards to prevent hallucinations and out-of-scope inquiries.
- **Edge-Case & Fallback Workflows:** Mapped error codes to step-by-step diagnostics and routed unresolvable issues to email support.

Intelli Global Corporation

Jun 2025 – May 2026

QA Engineer Intern (Part Time and Full Time)

Kwun Tong, Hong Kong

- **Data Integrity & Automation:** Engineered Python (NumPy) scripts to clean and validate large-scale datasets and raw CSVs for the project database.
- **Quality Assurance:** Reported and verified bugs; ran regression testing to ensure stability after feature deployments.
- **Documentation:** Managed system specs and user guides; standardized manuals to improve clarity.
- **System Deployment:** Deployed and configured openMAINT using Docker on Windows for smart building asset management.

Featured Projects

MTR Tracker Lite — *React Native, Expo, REST APIs* — Built a streamlined mobile app for real-time Hong Kong MTR tracking; integrated GPS, train APIs, and HKO weather data for a fast, ad-free experience.

Muse Dash Web Game — *Node.js, Express, Socket.io, Bcrypt* — Built both the frontend game client and a real-time backend with bidirectional event-based communication; implemented secure user authentication and password hashing.

Library Management System — *OOP, UML, Java* — Designed software architecture with use-case specifications for user roles, publishing pipelines, access control, and status tracking dashboards.

Fruit Ripeness Classification — *Python, Transfer Learning, EfficientNet, YOLO* — Trained an EfficientNet-B0 model to classify ripeness stages of three fruit types; used YOLO bounding boxes to segment inputs, achieving 93% test accuracy.

Tiny Smart Home — *ESP32, Home Assistant, IoT, Ollama AI* — Built a smart home system with ESP32 sensors (DHT11, MQ2, UV); visualized data on a dashboard, built a physical demo model, and implemented automation scripts.

Technical Skills

Languages: Python, C/C++, JavaScript/Node.js, HTML/CSS, Java

Spoken: Cantonese (4.5/5), English (4/5), Mandarin (4.5/5)

AI & ML: PyTorch, NumPy & Scikit-learn, Deep Learning (CNN, RNN), Computer Vision, GAN & Transformers, Reinforcement Learning

Backend & Systems: MySQL, Express.js & Socket.io, Linux, Object-Oriented Design

Tools & Hardware: Git, STM32/ESP32, MIPS Architecture, Hardware Communication Protocols (I2C, ADC), Docker